ENGINEERING DESIGN REPORT

Evolution, Cellular Planning, Spectrum Use\nand Interference Management

ECA56 – Advanced Wireless Communication Systems

CO1 Integrated Engineering Analysis and Planning Assignment

Report detail	Information
Course outcome	CO1 – Mobile-system evolution, cellular architecture, spectrum utilization and interference management
Assessment	Open-ended engineering design report
Planning scenario	Regional wireless operator – next-stage mobile network
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Executive Summary

This report evaluates a regional operator's network plan using the supplied geometry, traffic, mobility, interference and spectrum data. The planning area generates 80 Erlangs of busy-hour offered traffic. Erlang B analysis shows that 92 traffic channels are sufficient to keep blocking at or below 2% (approximately 1.922%), while the initial 100-channel group gives approximately 0.399% blocking under the stated pooled-trunk assumption.

A hexagonal cell of radius 2 km has an idealized area of 10.392 km^2; the 19-cell planning cluster covers approximately 197.45 km^2 geometrically. With reuse cluster size N = 7, the reuse fraction is 1/7 and the co-channel reuse distance is 9.165 km. The supplied interference case produces a co-channel C/I of 15 dB, but after adding adjacent-channel interference the calculated SINR falls to 13.81 dB, below the 15 dB target.

The recommended strategy is to retain low-band spectrum for broad-area coverage and mobility, deploy 3.5 GHz as the principal capacity layer in traffic hotspots, and use 28 GHz selectively for dense, line-of-sight high-capacity zones. Capacity should be expanded first through targeted sectorization and small-cell overlays, supported by measured traffic and radio-planning data. The calculations are first-order planning estimates and must be validated with field measurements, detailed propagation models, real busy-hour traffic distributions and regulatory spectrum conditions.

1. Requirements, Planning Inputs and Assumptions

The objective is to produce an integrated engineering recommendation connecting mobile-generation evolution with cell geometry, traffic capacity, mobility, interference, spectrum propagation and international coordination. The supplied assignment values are treated as the design basis.

Parameter	Supplied value
Cell model and radius	Hexagonal model; r = 2 km
Planning cluster	19 cells
Frequency reuse cluster size	N = 7
Subscribers and traffic	1,000 subscribers; 0.08 Erlang/subscriber in busy hour
Initial channel group	C = 100 traffic channels; blocking target ≤ 2%
Mobility	60 km/h; 6 s overlap dwell; 1 km nominal route cell length
Handoff and availability	0.5 s handoff interruption; target cell unavailable 1% of time
Interference	S = −70 dBm; I_1 = −85 dBm; I_2 = −90 dBm; target SINR = 15 dB
Spectrum comparison	900 MHz, 3.5 GHz, 28 GHz; d = 100 m; c = 3 × 10^8 m/s

1.1 Assumptions and interpretation

- The hexagonal model uses regular, equal-sized cells and idealized non-overlapping geometric tessellation. The stated radius is treated as the center-to-vertex radius of a regular hexagon.
- Erlang B assumes a loss system with Poisson call arrivals, exponentially distributed holding times, no queueing and a common pooled group of channels.
- The 1,000 subscribers are assumed to generate independent busy-hour traffic that can be statistically pooled into one logical trunk group. Actual geographic distribution may require several groups/sites.
- The 1 km route cell length is interpreted as the distance travelled between successive cell-boundary crossings along the route.
- For SINR, thermal noise is not supplied; therefore the requested calculation uses the desired signal divided by the sum of the two stated interference powers. The result is an interference-limited SINR estimate.
- The 1% target-cell unavailability is treated as the probability that a target cell cannot serve a handoff at the instant it is needed. This is a simplified per-handoff drop-risk estimate, not a full reliability model.

2. Stage 1 – Evolution of Mobile Communication Systems

Generation	Service capability	Access / radio technology	Key limitation and engineering response
1G	Analog voice	AMPS; 800 MHz; 30 kHz analog FDMA channels	Limited capacity, weak privacy and relatively poor spectral efficiency; motivated digital cellular systems.
2G	Digital voice, SMS and basic data	GSM at 900 MHz; 200 kHz carrier with 8 TDMA time slots	Much improved capacity, security and service features; data rates and multimedia capability remained limited.
3G	Mobile internet, packet data and multimedia	UMTS; 5 MHz carrier	Higher data capability than 2G, but capacity and spectral efficiency could not meet later broadband demand economically.
4G	Mobile broadband, IP services and high data rates	LTE; OFDMA downlink, SC-FDMA uplink and MIMO	Improved spectral efficiency and broadband capacity; still challenged by very high density, ultra-low latency and massive device counts.
5G	eMBB, URLLC and massive IoT	5G NR; sub-6 GHz/mid-band and mmWave; beamforming and massive MIMO	Enables flexible services and high capacity, but mmWave coverage, blockage, site density and deployment cost are challenging.
6G vision	Immersive/holographic communication, tactile internet, advanced sensing and integrated connectivity	Research vision includes THz spectrum, AI-native networking and ISAC	Not yet a mature, universally deployed generation; faces propagation, hardware, energy, security and standardization challenges.

2.1 1G wavelength and idealized channel count

Wavelength at 800 MHz

$\lambda = c / f = (3 \times 10^8 \text{ m/s}) / (800 \times 10^6 \text{ Hz})$

Result: $\lambda = 0.375$ m (37.5 cm).

This wavelength is relevant to antenna dimensions and propagation behavior in the 800 MHz band.

Idealized number of 30 kHz channels in 10 MHz

$N = 10 \text{ MHz} / 30 \text{ kHz} = 10{,}000 \text{ kHz} / 30 \text{ kHz}$

Result: N = 333.33, or 333 complete channel-widths.

This is only a gross arithmetic upper bound. It ignores guard bands, duplex separation, channelization rules, control channels, unusable edge spectrum and practical allocation constraints. If the 10 MHz is a paired allocation, uplink/downlink partitioning further reduces the number available in each direction.

The move from 1G analog FDMA to 2G digital transmission and TDMA allowed multiple users to share a carrier in separate time slots, improved spectral efficiency, enabled digital encryption and error-control techniques, and supported services such as SMS. Across generations, spectrum planning evolved from fixed narrowband voice channels toward flexible wideband carriers, spatial reuse, adaptive scheduling, beamforming and more sophisticated interference coordination.

3. Stage 2 – Cell Coverage and Frequency Reuse

Area of one regular hexagonal cell

$A_{cell} = (3\sqrt{3} / 2)\, r^2 = (3\sqrt{3} / 2)(2 \text{ km})^2$

Result: Acell = 10.392 km^2.
This is the geometric area of one ideal regular hexagon when r is the center-to-vertex distance.

Area of 19-cell cluster

Acluster = 19 × 10.392 km^2
Result: Acluster = 197.45 km^2.
The cluster area is a geometric planning estimate; real site coverage will be irregular and may overlap or leave coverage gaps.

Frequency reuse fraction

Reuse fraction = 1 / N = 1 / 7
Result: 1/7 = 0.1429 (14.29%).
In an ideal equal allocation, each cell receives one-seventh of the total system channel set. The same frequencies are reused in other clusters.

Co-channel reuse distance

$D = r\sqrt{(3N)} = 2\text{ km} \times \sqrt{21}$
Result: D = 9.165 km.
This is the ideal center-to-center spacing between co-channel cells for the conventional hexagonal model.

3.1 Model limitations: geometry versus radio coverage

The hexagonal approximation assumes uniform terrain, equal cell radii, regular site spacing, omnidirectional idealized cells and a consistent propagation environment. Real radio coverage depends on antenna height and pattern, transmit power, frequency, clutter, terrain, building penetration, shadowing, fading, interference and receiver sensitivity. A geometric cell boundary therefore does not guarantee usable signal quality or service at every point.

3.2 Capacity-expansion options and recommendation

Technique	Coverage/capacity effect	Interference and handoff implications
Cell splitting	Divides a large cell into smaller cells with lower-radius coverage; increases spatial reuse and capacity.	More sites and boundaries; can increase handoff rate and planning complexity. Requires careful power control and frequency planning.
Sectoring	Uses directional antennas (commonly 3 sectors of 120°) to reduce the angular region served by each site.	Can reduce received co-channel interference and improve reuse; sector boundaries can add handoff events.
Microcells	Small outdoor cells for streets, campuses and dense traffic areas.	Adds localized capacity; more frequent handoffs and tighter neighbor planning.
Picocells	Very small cells for indoor venues, stations and enterprise spaces.	Improves indoor capacity; adds mobility-management and interference-coordination requirements.
Femtocells	Low-power user/enterprise premises cells, often for indoor coverage.	Improves indoor service but may create access, synchronization and cross-tier interference issues.

Recommended measure: use sectorization at existing high-demand macro sites as the first targeted expansion, then add microcells at verified hotspots. Sectorization can improve spatial isolation without immediately multiplying the number of sites. Where demand remains concentrated, microcells add substantial localized capacity. This staged approach balances cost and capacity, while recognizing that microcells and smaller coverage footprints increase handoff frequency. A drive-test and traffic heat map should determine exact locations and sector azimuths.

4. Stage 3 – Traffic and Channel Capacity

Offered busy-hour traffic

A = subscribers × traffic per subscriber = 1,000 × 0.08 Erlang

Result: A = 80.0 Erlangs.

An Erlang is one circuit occupied continuously for one hour. The result represents aggregate offered traffic during the busy hour, not the number of simultaneous subscribers at every instant.

4.1 Erlang B model

For C channels and offered traffic A, Erlang B gives the probability that a new call is blocked because all channels are occupied. The recursive calculation used is:

$$B(0,A) = 1; \quad B(C,A) = [A \cdot B(C-1,A)] / [C + A \cdot B(C-1,A)]$$

Blocking with the initial 100 channels

B(100, 80) = ErlangB(A = 80, C = 100)

Result: B = 0.003992 = 0.399%.

The initial 100-channel group meets the ≤ 2% blocking requirement with a margin of approximately 1.601 percentage points.

Minimum channel count for ≤ 2% blocking

Evaluate Erlang B for increasing integer C; B(91,80) = 2.254% and B(92,80) = 1.922%

Result: Cmin = 92 channels; blocking = 1.922%.

Since 91 channels fail the target and 92 channels meet it, 92 is the smallest channel count under the Erlang B assumptions.

Number of 100-channel base-station channel groups

Groups = ceiling(92 / 100)

Result: 1 group.

One 100-channel group is sufficient in aggregate. This is not a complete site count: spectrum availability, duplexing, sector allocation, control channels, traffic geography, trunking boundaries, coverage and capacity per carrier/site are not specified.

4.2 Capacity, blocking and infrastructure trade-offs

Increasing channel count lowers blocking but consumes more spectrum or requires additional carriers and radio equipment. The initial 100-channel group already provides better than the 2% target for the pooled 80-Erlang load, so adding another full 100-channel group solely to meet this mathematical target would be inefficient. The operator should instead validate the distribution of traffic by site and sector. A site with a concentrated load may block even when the network-wide aggregate appears adequate.

The offered-traffic estimate assumes all subscribers have the same busy-hour traffic intensity and that calls can be pooled into a common loss system. Real traffic is time-varying and spatially uneven; call holding-time distributions, data traffic, priority classes and handoff reservations can also change blocking. Erlang B is a voice-circuit model and is only an approximate planning analogue for packet-based 4G/5G systems.

5. Stage 4 – Mobility and Handoff

Distance travelled during 6 s overlap

Speed = 60 km/h = 16.667 m/s; distance = 16.667 × 6

Result: Distance = 100.0 m.

The mobile travels about 100 m while it remains in the overlap zone. The network has this short interval to detect, prepare and complete a handoff.

Estimated hard handoffs per minute

60 km/h = 1 km/min; one route cell length = 1 km

Result: Approximately 1.0 hard handoff per minute.

This estimate assumes one cell boundary is crossed for each 1 km of travel and ignores route geometry, cell overlap, ping-pong handoffs and irregular cell shapes.

5.1 Hard, soft and softer handoff

Handoff type	Connection continuity	Resource use and typical context
Hard handoff	Break-before-make: the old radio link is released before or as the new one is established. A brief interruption can occur.	Usually lower simultaneous radio-resource demand; common in GSM and LTE-style inter-frequency/intra-frequency procedures depending on system.
Soft handoff	Make-before-break: a mobile can communicate with multiple base stations/cells during transition.	Improves continuity and reduces edge failures, but consumes resources in multiple cells; historically associated with CDMA systems.
Softer handoff	Make-before-break transition between sectors of the same base station.	Combines sector signals/resources within one site; improves continuity but requires coordination and additional radio resources.

Simple per-handoff call-drop risk estimate

P(drop per handoff) ≈ P(target unavailable) = 1%

Result: Approximately 1% per handoff under the stated simplifying assumption.

The 0.5 s interruption is a separate service-quality exposure: it is 0.5/6 = 8.33% of the 6 s overlap dwell interval. It does not, by itself, imply an 8.33% drop probability. The 1% estimate assumes target unavailability is independent of the handoff event and that no alternate target, retry, buffering or make-before-break mechanism rescues the call.

The simple estimate omits radio-link failure probability, handoff preparation failures, signaling delay, congestion in the target cell, correlated outages, fading and the number of handoffs during a call. A deployment-grade drop model should combine these probabilities and use measured handoff success statistics.

6. Stage 5 – Interference and Spectrum-Band Comparison

6.1 Co-channel C/I

Carrier-to-interference ratio for co-channel interferer alone

$C/I = S - I_1 = (-70\ \text{dBm}) - (-85\ \text{dBm})$

Result: C/I = 15.0 dB.

With only the co-channel interferer, the ratio equals the 15 dB minimum target exactly. There is no margin for additional interference or noise.

6.2 Combine co-channel and adjacent-channel interference

Decibel powers must not be added directly. Convert each interference power to a linear ratio relative to 1 mW, add the powers, and convert the sum back to dBm.

Total interference power

I_total(mW) = 10^(−85/10) + 10^(−90/10)

Result: I_total = -83.807 dBm.

The combined interference is greater than either interferer alone; the adjacent-channel contribution raises the total interference above −85 dBm.

SINR using supplied desired signal

SINR = S − I_total = (−70 dBm) − (−83.807 dBm)

Result: SINR = 13.81 dB.

The 15 dB target is NOT met. The shortfall is approximately 1.19 dB. Thermal noise is excluded because no noise power or receiver bandwidth is supplied; including noise would reduce SINR further.

6.3 Mitigation recommendation

Apply coordinated interference management using improved filtering and channel assignment: increase adjacent-channel isolation with suitable RF filters and guard-band/channel planning, while coordinating reuse and antenna azimuth/tilt to reduce co-channel coupling. In a 5G layer, beamforming and power control can further limit interference toward unintended receivers. The trade-off is that guard bands reduce usable spectrum, tighter filtering increases RF cost and insertion loss, and power reduction or directional scheduling may reduce coverage or scheduling flexibility.

6.4 Wavelength and free-space path loss at 100 m

Use $\lambda = c/f$ and $FSPL = 20 \log_{10}(4\pi d/\lambda)$, with $c = 3 \times 10^8$ m/s and d = 100 m.

Frequency	Wavelength	FSPL at 100 m	Difference vs 900 MHz
0.9 GHz	0.33333 m	71.53 dB	+0.00 dB
3.5 GHz	0.08571 m	83.32 dB	+11.80 dB
28 GHz	0.01071 m	101.38 dB	+29.86 dB

The free-space calculation gives approximately 71.53 dB at 900 MHz, 83.32 dB at 3.5 GHz and 101.38 dB at 28 GHz. At the same distance, 3.5 GHz has about 11.80 dB more free-space loss than 900 MHz, while 28 GHz has about 29.86 dB more.

Band	Likely propagation / penetration	Antenna and deployment implications
900 MHz	Generally better diffraction and building penetration; broad-area and indoor coverage are more practical.	Longer wavelength (33.3 cm) means larger antenna elements; fewer sites can cover a broad area, but available bandwidth may be limited.
3.5 GHz	Intermediate propagation and penetration; useful balance of coverage and capacity in urban/suburban macro layers.	Wavelength about 8.57 cm supports compact arrays and massive MIMO; moderate site density and beamforming can deliver high capacity.
28 GHz	Higher free-space loss and stronger sensitivity to blockage, foliage and building penetration; often needs line-of-sight or carefully planned reflections.	Short wavelength (1.07 cm) enables compact antenna arrays and narrow beams; dense small-cell placement is often needed for continuity and capacity.

Higher free-space path loss alone does not determine achievable performance. Link budget also depends on antenna gain, array size, beamforming gain, transmit power limits, receiver noise figure, bandwidth, coding, fading, blockage, atmospheric/rain effects and deployment density. A higher-frequency link can use larger bandwidth and directional array gain to achieve high peak capacity, but coverage and mobility are more challenging.

7. Stage 6 – International Spectrum Coordination

Organization / forum	Role in spectrum coordination
ITU-R	The ITU Radiocommunication Sector develops technical studies, recommendations and regulatory material to support efficient and harmonized use of radio-frequency spectrum and satellite orbits.

World Radiocommunication Conference (WRC)	A periodic treaty-level conference of ITU Member States that reviews and revises the Radio Regulations, including allocations, service conditions and international procedures.
Radiocommunication Bureau (BR)	The ITU-R secretariat/administrative bureau that supports conferences and study groups and administers international frequency coordination, notification, recording and related procedures.

International coordination establishes a common framework for spectrum allocations and cross-border compatibility. National regulators use the Radio Regulations and WRC outcomes when developing national allocation tables, licensing rules and auction plans. Operators must then deploy within national licences and technical conditions, while coordinating with neighboring countries where signals may cross borders. Harmonization can improve equipment economies of scale and roaming compatibility, but national regulators still decide domestic licensing and implementation details.

8. Integrated Deployment Recommendation

8.1 Recommended phased strategy

Phase	Network action	Engineering justification and checks
Phase 1 – Stabilize coverage	Retain/optimize the 900 MHz layer for wide-area coverage, indoor reach and mobility; audit existing sites, antenna patterns and coverage holes.	The 900 MHz wavelength is 0.333 m and its 100 m FSPL is the lowest of the compared bands. Validate with drive tests, indoor tests and coverage prediction.
Phase 2 – Add targeted capacity	Use 3.5 GHz as the primary capacity layer in towns, transport corridors and high-traffic zones; add sectors or microcells where measured demand requires them.	The 3.5 GHz band offers a balance of coverage and capacity. The calculated 80 Erlangs require at least 92 pooled channels for ≤2% blocking; the existing 100-channel group meets the aggregate target, but local sector loads must be checked.
Phase 3 – Densify high-capacity hotspots	Introduce 28 GHz small cells at dense venues or fixed high-demand zones where line-of-sight and backhaul are available.	28 GHz has about 29.86 dB more free-space loss than 900 MHz at 100 m, so use dense placement, beamforming and robust fallback to lower bands.
Phase 4 – Optimize mobility and interference	Tune handoff thresholds, neighbor lists, antenna tilt/azimuth, power, filtering and channel assignments; use coordinated beamforming where supported.	A 60 km/h user travels about 100 m in the 6 s overlap zone and crosses a nominal 1 km route cell about once per minute. The interference example yields 13.81 dB SINR, below target, so mitigation and margin are needed.
Phase 5 – Prepare for evolution	Keep architecture and spectrum strategy flexible for 5G-Advanced and future 6G capabilities, including AI-assisted optimization and sensing/communications convergence as standards mature.	6G is a vision rather than an immediate replacement. Pilot new capabilities only after standards, devices, spectrum policy, security and business cases are established.

8.2 Key assumptions and uncertainties to validate before deployment

- Confirm whether the stated 10 MHz allocation is paired or unpaired, and account for guard bands, duplex spacing, control channels and regulatory channelization.
- Measure busy-hour voice and data demand per site, sector and service class; verify whether the 80-Erlang aggregate load can genuinely be pooled.
- Confirm the exact definition of cell radius and route cell length, and replace ideal hexagons with GIS-based site locations and clutter-aware propagation predictions.

- Measure real handoff success, interruption, ping-pong rate, target-cell congestion and outage correlation at 60 km/h and other mobility conditions.
- Measure receiver noise, bandwidth, RF filtering, adjacent-channel leakage and real interference distributions; the supplied SINR calculation excludes thermal noise.
- Validate band-specific link budgets with antenna gains, transmit power limits, body/building losses, foliage, rain, blockage and backhaul availability.
- Check national spectrum licensing, WRC/ITU-R constraints, cross-border coordination, equipment availability, site permits, power and energy costs.
- Evaluate resilience, cybersecurity, emergency-service continuity, synchronization and operational support for dense multi-layer deployments.

9. Conclusion

The analysis supports a layered and phased regional network rather than a single-band solution. The evolution from 1G analog FDMA to digital 2G TDMA, 3G broadband, 4G OFDMA/MIMO and 5G NR demonstrates the increasing importance of spectral efficiency, spatial reuse, flexible scheduling and interference control. The 6G vision extends these trends but should be treated as a research and standards roadmap.

The idealized 19-cell cluster covers 197.45 km^2, with a reuse distance of 9.17 km for N = 7. The aggregate busy-hour load is 80 Erlangs; 92 channels are the calculated minimum for a 2% blocking limit, and the initial 100-channel group yields only 0.399% blocking in the pooled model. Mobility analysis indicates about 100 m of travel during the overlap interval and roughly one hard handoff per minute along the assumed route. The interference case is more critical: combining co-channel and adjacent-channel power gives an estimated SINR of 13.81 dB, which fails the 15 dB target.

Accordingly, the operator should use 900 MHz for broad coverage, 3.5 GHz for the main capacity layer and 28 GHz selectively for dense hotspots. Sectorization and measured microcell additions should precede broad densification. The final design must be confirmed through field measurements, detailed link budgets, per-sector traffic forecasts and national spectrum authorization. This recommendation connects capacity, mobility, propagation and interference constraints while retaining flexibility for future network generations.

References

- [1] ITU-R, Recommendation M.2083-0, “IMT Vision – Framework and overall objectives of the future development of IMT for 2020 and beyond,” September 2015.
- [2] ITU-R, Recommendation M.2160-0, “Framework and overall objectives of the future development of IMT for 2030 and beyond,” November 2023.
- [3] ITU, Radio Regulations, current edition and applicable WRC Final Acts. International frequency allocation and coordination framework.
- [4] 3GPP, TS 38.300, “NR; NR and NG-RAN Overall Description,” applicable release.
- [5] T. S. Rappaport, Wireless Communications: Principles and Practice, 2nd ed., Prentice Hall, 2002.
- [6] W. C. Y. Lee, Mobile Cellular Telecommunications: Analog and Digital Systems, 2nd ed., McGraw-Hill, 1995.

Note: Numerical calculations in this report are derived directly from the planning data supplied in the assignment. The references above provide supporting technical background; the report does not assume that illustrative planning values are regulatory allocations or final site-design parameters.

Appendix A – Numerical Results at a Glance

Item	Calculated result
Wavelength at 800 MHz	0.375 m
Ideal 30 kHz channel-width count in 10 MHz	333 complete channel-widths; ignores practical constraints
One hexagonal cell area	10.392 km^2
19-cell cluster area	197.45 km^2
Reuse fraction (N = 7)	14.29%
Co-channel reuse distance	9.165 km
Offered busy-hour traffic	80 Erlangs
Erlang B at 100 channels	0.399%
Minimum channels for ≤2%	92 channels (1.922% blocking)
100-channel groups required	1
Distance in 6 s overlap	100 m
Estimated hard handoffs	1 per minute
Co-channel C/I	15 dB
Combined interference	-83.81 dBm
SINR (noise excluded)	13.81 dB; target not met
FSPL at 900 MHz / 3.5 GHz / 28 GHz	71.53 / 83.32 / 101.38 dB